Ghost Layer / QuantGPUMgment

Ideology Validation & Consolidated Change Report

VERDICT: The core business thesis is sound and well-argued. The evidence layer supporting it is not yet trustworthy — the one real end-to-end training test on file contradicts the numbers in every client-facing report, and those reports were built from hardcoded constants and simulated data, not measurements. Fix the evidence before showing this to a client.

1. What the Ideology Gets Right

The underlying business thesis, as laid out in the business-plan documents, holds up on its own merits:

- **The inefficiency is real and well-documented.** Low GPU utilization during training is a widely cited industry problem, and mixed precision / gradient checkpointing / FlashAttention are genuine, published techniques — none of this is invented.
- **Performance-fee pricing is the right model for this category.** Getting paid only on verified savings removes the client's biggest objection ("prove it works before I pay") and mirrors a model with real precedent (MosaicML built a venture-backed, later-acquired company on a similar thesis).
- **The phased-trust rollout is correct and should not be shortcut.** Read-only observation → narrow auto-apply → broader automation is the right sequencing given that a bad automated change can silently corrupt a client's model.
- **Zero-capital framing for the MVP is defensible.** The agent runs inside the client's own environment, and the core techniques are free and open-source. This isn't a capital-intensive business to start.
- **Naming a correctness-verification layer as mandatory, not optional, is the right call** — and, as covered in Section 2 below, it's also the one thing in this project that is currently working exactly as designed.

2. Critical Findings — Evidence Doesn't Support the Pitch Yet

These are the issues that matter before any of this reaches a client. Each is drawn directly from the files in the project.

P0 — CRITICAL **2.1 The only real training test on file contradicts every client-facing report.**

Every polished audit report in this project (*audit_report.md*, *client_audit_report.md*, *gameplan_audit_report.md*, *ghost_layer_audit_receipt.md*) shows the same story: a clean +32% to +58% speedup, "Optimization Verified & Audit Complete," money earned. But *real_training_validation_report.md* — generated from an actual PyTorch training loop that actually ran (*examples/test_real_training_loop.py*), not synthetic input — shows the optimized run was **45.9% slower** than baseline, with **negative** dollar savings, and the correctness verifier **rejected** the change outright: divergence hit 0.677 against a safety threshold of 0.25, so auto-apply was correctly revoked.

This is not a minor discrepancy. It means the one time this system was actually tested end-to-end on real code, it failed the exact test the business plan says must never be skipped — and none of the other reports disclose that this happened. If a prospective client saw both files side by side, the pitch would not survive the conversation.

Source: real_training_validation_report.md vs. audit_report.md / client_audit_report.md / gameplan_audit_report.md / ghost_layer_audit_receipt.md

P0 — CRITICAL 2.2 The headline speedup numbers are hardcoded constants, not measurements.

In *ghost_layer/decision/engine.py*, every recommendation carries a fixed `estimated_speedup_pct`: mixed precision is always 45.0%, DataLoader tuning is always 25.0%, FlashAttention is always 35.0%, gradient checkpointing is always 20.0%, batch scaling is always 30.0%. These are the same numbers in every report regardless of the client's actual hardware, model, or measured before/after result — because they are rule-level literals, not outputs of a measurement.

Presented in a document titled "Optimization Verified & Audit Complete" with a client's name on it, this reads as a measured result. It is currently a published industry range attached to a template.

Source: ghost_layer/decision/engine.py

P1 — HIGH 2.3 Client-facing "audit reports" are generated from simulated data, not client telemetry.

The reports in this bundle trace back to *ghost_simulation_db.json* and a 250-run synthetic benchmark set (*ghost_sim_test_250_kb.json*, *SIMULATED_250_LLM_GPU_BENCHMARKS.md*), generated by scripts in *scratch/*. Nothing marks the client-facing versions as demo output — they use the same "Client: Client AI Research Lab" / "Status: Optimization Verified" template a real engagement would produce.

Sending a demo-data report to a real prospect without a visible "illustrative / simulated" label is a misrepresentation risk, independent of how good the underlying technique is.

Source: ghost_simulation_db.json, ghost_sim_test_250_kb.json, SIMULATED_250_LLM_GPU_BENCHMARKS.md, scratch/generate_*.py

P1 — HIGH 2.4 Telemetry math produces impossible values.

"DataLoader I/O Stall Percentage" is reported at **5287.9%** in *gameplan_audit_report.md*, **1173.0%** in *ghost_layer_audit_receipt.md*, and **524.5%** in *real_training_validation_report.md*. A stall percentage cannot exceed 100% — this points to a unit or denominator bug in *ghost_layer/telemetry/watcher.py*, not a hardware condition. Any report containing this number as-is would visibly undermine the tool's credibility to a technical reader within seconds.

Source: gameplan_audit_report.md, ghost_layer_audit_receipt.md, real_training_validation_report.md, ghost_layer/telemetry/watcher.py

P2 — MEDIUM 2.5 GTM language overreaches in ways that will cost credibility with the exact buyers targeted.

GHOST_LAYER_REALITY_CHECK_PLAN.md frames replacing $35,000/month principal CUDA/MLOps engineers with "$0/month" because "the founders + Agentic AI" will do the work. A Series A AI startup evaluating whether to let this system touch their training run will read that line as a red flag about who is actually behind the verification claims, not as a cost-savings pitch. The main business-plan document (*quant-ai-training-efficiency-business-plan.md*) is considerably more measured and cites real, attributable sources (GPU utilization survey data, NVIDIA DSX, MosaicML) — that version is the one worth standardizing on.

Similarly, "Trojan Horse" as an internal label for the free-trial trust-building strategy (*GHOST_LAYER_REALITY_CHECK_PLAN.md*, Section 3) is fine as private shorthand but should never appear in anything a client, investor, or contractor might see.

Source: GHOST_LAYER_REALITY_CHECK_PLAN.md vs. quant-ai-training-efficiency-business-plan.md

3. Consolidated & Tightened Recommended Changes

The project contains five separate audit reports and several planning docs, each proposing overlapping fixes. Below is that list deduplicated, reconciled where they conflicted, and ordered by what actually blocks this from being shown to a client.

P0 — FIX BEFORE ANY CLIENT SEES THIS

Change	Why / where
Investigate and resolve the correctness-verifier rejection in the real training run before making any auto-apply claim to a client.	real_training_validation_report.md is the only genuine evidence of end-to-end behavior, and it failed.
Stop generating client-named "Audit Report" documents from simulated/synthetic data. Add a visible SIMULATED/DEMO watermark to any report not produced from a live client run, and keep the two templates visually distinct.	ghost_simulation_db.json, ghost_sim_test_250_kb.json, SIMULATED_250_LLM_GPU_BENCHMARKS.md
Replace hardcoded per-rule speedup percentages with either (a) a measured before/after number from the actual run, or (b) an explicitly labeled "typical published range" — never presented as a verified result.	ghost_layer/decision/engine.py
Fix the DataLoader stall-percentage calculation so it cannot exceed 100%; audit the denominator in the telemetry watcher.	ghost_layer/telemetry/watcher.py

P1 — FIX BEFORE ACTIVE OUTREACH

Change	Why / where
Standardize all outward-facing materials on the measured, source-cited version of the business plan (quant-ai-training-efficiency-business-plan.md); retire the "$0/month engineering" and "Trojan Horse" framing from anything client- or investor-facing.	GHOST_LAYER_REALITY_CHECK_PLAN.md vs. the main business-plan doc
Require every "Auto-apply Safe: Yes" label to be backed by a passed correctness-verifier run, not just the rule heuristic that currently sets it.	ghost_layer/decision/engine.py, ghost_layer/verification/verifier.py

Change	Why / where
Define and document the baseline-measurement methodology in the client contract before any engagement starts (the business plan already flags this risk in Section 7 — it needs to move from "risk noted" to "contract clause drafted").	quant-ai-training-efficiency-business-plan.md, Section 7 & 8
Run the real-training validation suite (examples/test_real_training_loop.py) against at least 2-3 different reference architectures before claiming any specific speedup range in a pitch deck.	examples/test_real_training_loop.py, examples/train_llm_gameplan_demo.py

P2 — TIGHTEN WHEN CONVENIENT

Change	Why / where
Consolidate the five near-duplicate audit-report files and multiple business-plan drafts (two versions of the same 26-31KB plan) into one canonical version each; keep dated snapshots elsewhere if history matters.	quant-ai-training-efficiency-business-plan.md vs. "...(1).md"; audit_report.md / client_audit_report.md / gameplan_audit_report.md / ghost_layer_audit_receipt.md
Keep the phased-trust GTM structure (free trial → read-only report → narrow auto-apply) but drop framing that describes the free trial primarily as a trust-extraction tactic; describe it as what it is — a genuine no-cost pilot.	GHOST_LAYER_REALITY_CHECK_PLAN.md, Section 3
Version-stamp the CLI/engine output (currently "v0.1.0" across all reports) so a client can tell which reports came from which engine revision once the fixes above ship.	ghost_layer/reporting/generator.py

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Bottom line: the business idea does not need to change. The evidence pipeline behind it does — right now the polished reports and the one real test tell opposite stories, and that gap is the only thing standing between this and a credible pilot pitch.